

AWS Migration Business Case

Prepared for Customer

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Executive Overview

This comprehensive business case analysis presents the strategic and financial benefits of migrating to Amazon Web Services (AWS). Based on detailed assessment of your current infrastructure and business requirements, this analysis demonstrates significant cost savings, operational improvements, and competitive advantages achievable through cloud migration.

Financial Summary

Migration Strategy	Current Annual Cost	AWS Annual Cost	Annual Savings	5-Year Savings	Timeline
Current State	\$0.3M	-	-	-	-
Lift Shift	\$0.3M	\$0.3M	\$0.1M	\$0.4M	6 months
Replatform	\$0.3M	\$0.2M	\$0.1M	\$0.7M	12 months
Refactor	\$0.3M	\$0.2M	\$0.2M	\$0.9M	18 months
Cloud Native	\$0.3M	\$0.1M	\$0.2M	\$1.1M	24 months

Key Recommendations

- Proceed with Cloud Native migration strategy for maximum ROI
- Implement migration in phases over 24 months
- Prioritize security and compliance framework setup
- Invest in team training and change management
- Establish AWS Landing Zone for governance

1. Total Cost of Ownership Analysis

TCO Analysis Based on Customer Report Data ## Current State Analysis (from report.pdf) - **Current Annual Infrastructure Cost**: \$0.3M - **Server Count**: 368 servers - **Application Count**: 0 applications - **Current Monthly Cost**: \$0.0M ## Migration Scenario Analysis ### Lift Shift Migration - **Complexity**: Low - **Timeline**: 6 months - **AWS Annual Cost**: \$0.3M - **Annual Savings**: \$0.1M - **5-Year Savings**: \$0.4M - **ROI**: 83% ### Replatform Migration - **Complexity**: Medium - **Timeline**: 12 months - **AWS Annual Cost**: \$0.2M - **Annual Savings**: \$0.1M - **5-Year Savings**: \$0.7M - **ROI**: 167% ### Refactor Migration - **Complexity**: High - **Timeline**: 18 months - **AWS Annual Cost**: \$0.2M - **Annual Savings**: \$0.2M - **5-Year Savings**: \$0.9M - **ROI**: 306% ### Cloud Native Migration - **Complexity**: Very High - **Timeline**: 24 months - **AWS Annual Cost**: \$0.1M - **Annual Savings**: \$0.2M - **5-Year Savings**: \$1.1M - **ROI**: 464% ## Recommended Strategy: Cloud Native - **Total 5-Year Savings**: \$1.1M - **Implementation Timeline**: 24 months - **Complexity Level**: Very High ## AWS Calculator Links - Lift & Shift: <https://calculator.aws/#/estimate?id=lift-shift-3> - Re-platform: <https://calculator.aws/#/estimate?id=replatform-3> - Modernization: <https://calculator.aws/#/estimate?id=modernize-3> *Analysis based on actual customer data from report.pdf*

2. Migration Strategy & Approach

Based on the provided information, I'll generate a migration scenario matrix for AnyCompany's cloud migration. Please note that some assumptions have been made due to limited details in certain areas. Migration Scenario Matrix: 1.

Rehost (Lift and Shift) Strategy: - Move existing applications and infrastructure to AWS with minimal changes - Utilize AWS EC2 instances to replicate on-premises servers Effort: Medium (3000-4000 person-hours) Timeline: 6-8 months Risk: Medium - Pros: Fastest migration path, minimal code changes - Cons: Doesn't fully leverage cloud benefits, may not solve scalability issues 2.

Replatform (Lift, Tinker, and Shift) Strategy: - Move applications to AWS while making cloud-optimized adjustments - Implement auto-scaling for web and application tiers - Migrate databases to Amazon RDS for improved management Effort: High (5000-6000 person-hours) Timeline: 10-12 months Risk: Medium - Pros: Improved performance and manageability, some cloud benefits - Cons: Requires more effort than rehost, potential application changes 3.

Refactor (Re-architect) Strategy: - Redesign core applications using cloud-native services - Implement microservices architecture for improved scalability - Utilize serverless computing where applicable (e.g., AWS Lambda) Effort: Very High (8000-10000 person-hours) Timeline: 18-24 months Risk: High - Pros: Full cloud benefits, addresses scalability and performance issues - Cons: Longest timeline, highest cost, significant application changes Recommended Phased Approach: Phase 1: Rehost (3-4 months) - Migrate non-critical systems and static content to AWS - Implement basic monitoring and security controls Effort: 1500-2000 person-hours Phase 2: Replatform (4-5 months) - Migrate and optimize core e-commerce platform - Implement auto-scaling for web and application tiers - Migrate databases to Amazon RDS Effort: 2500-3000 person-hours Phase 3: Refactor (6-8 months) - Re-architect inventory management system for real-time updates - Implement microservices for product catalog and personalization - Develop serverless functions for specific backend processes Effort: 4000-5000 person-hours Total Effort: 8000-10000 person-hours Total Timeline: 13-17 months Risk Mitigation Strategies: 1.

Implement robust testing and rollback procedures 2. Use blue-green deployment for critical systems 3. Conduct thorough staff training on cloud technologies 4. Establish a cloud center of excellence for best practices 5. Engage AWS support for complex migration challenges Dependency-aware Sequencing: 1. Start with static content and non-critical systems 2. Migrate databases before dependent applications 3. Implement new cloud-native services in parallel with existing systems 4.

Gradually shift traffic to new cloud infrastructure 5. Decommission on-premises systems as cloud replacements are validated This phased approach balances the need for quick wins (rehost) with the long-term benefits of cloud-native architecture (refactor). It allows AnyCompany to address immediate scalability concerns while progressively modernizing their application portfolio.

The approach also provides opportunities to build team expertise and adjust strategies based on lessons learned in earlier phases.

3. Risk Assessment & Mitigation

Given the migration readiness score of 3.2/5.0 and the financial analysis previously provided, I'll offer a more nuanced recommendation that takes into account both the organization's readiness and the financial implications of each migration scenario. Recommendation: Considering the migration readiness score of 3.2/5.0, which indicates moderate readiness with room for improvement, and the financial analysis of the four migration scenarios, I recommend a phased approach:

- 1. Initial Phase: Replatform Migration** Start with the Replatform migration scenario for the following reasons:
 - Moderate complexity aligns well with the current readiness level (3.2/5.0)
 - Balanced financial benefits: - Implementation Costs: \$75,000 - Annual Savings: \$96,377 - NPV (5-year): \$239,367 - IRR (5-year): 116.41% - Payback Period: 9.34 months
 - Lower risk compared to more complex scenarios
 - Provides a good foundation for future improvementsThis approach allows the organization to:
 - Gain experience with cloud migration
 - Realize significant cost savings and ROI
 - Improve cloud readiness for future phases
- 2. Mid-term Phase: Refactor Migration** As the organization's cloud maturity improves, transition to the Refactor migration scenario:
 - Higher complexity but greater benefits: - Implementation Costs: \$120,000 - Annual Savings: \$147,283 - NPV (5-year): \$390,855 - IRR (5-year): 116.41%
 - Allows for modernization of key applications
 - Prepares the ground for more advanced cloud-native architectures
- 3. Long-term Vision: Cloud Native Migration** Set Cloud Native migration as the long-term goal:
 - Highest NPV: \$483,343 - Maximum annual savings: \$181,220
 - Requires highest level of cloud maturity and readiness**Implementation Plan:**
 - 1. Immediate Actions (0-3 months):**
 - Address gaps identified in the migration readiness assessment
 - Develop a detailed plan for the Replatform migration
 - Begin staff training and skill development
 - 2. Replatform Migration (3-15 months):**
 - Execute the Replatform migration over 12 months
 - Continuously monitor and optimize migrated workloads
 - Reassess migration readiness and cloud maturity
 - 3. Transition to Refactor (15-18 months):**
 - Plan the transition from Replatform to Refactor
 - Identify key applications for refactoring
 - Enhance staff skills for more complex cloud architectures
 - 4. Refactor Migration (18-36 months):**
 - Execute the Refactor migration over 18 months
 - Continue optimization and maturity improvement
 - 5. Prepare for Cloud Native (36-42 months):**
 - Assess readiness for Cloud Native migration
 - Develop a strategy for transitioning to fully cloud-native architectures
 - 6. Cloud Native Migration (42+ months):**
 - Begin Cloud Native migration for suitable workloads
 - Continue to evolve and optimize cloud infrastructureThis phased approach allows the organization to:
 1. Match migration complexity with current readiness
 2. Realize incremental benefits and ROI
 3. Continuously improve cloud maturity and readiness
 4. Minimize risk while maximizing long-term benefitsBy following this plan, the organization can navigate from its current moderate readiness (3.2/5.0) to a high level of cloud maturity, ultimately achieving the full benefits of cloud-native architecture while managing risks and ensuring sustainable progress.

4. Security & Compliance Framework

Thank you for providing additional context about the compliance requirements. Based on this information and the previously provided details about AnyCompany, I'll outline a comprehensive security architecture focusing on PCI DSS and SOX compliance for a retail organization with unknown locations. Please note that this is a general approach and may need to be adjusted based on specific company details. Security Architecture for AnyCompany: 1.

Compliance Requirements: a) PCI DSS (Payment Card Industry Data Security Standard): - Secure cardholder data - Maintain a vulnerability management program - Implement strong access control measures - Regularly monitor and test networks - Maintain an information security policy b) SOX (Sarbanes-Oxley Act): - Ensure accuracy of financial reporting - Maintain effective internal controls - Provide accountability for corporate disclosures 2.

Security Controls: a) Network Security: - Implement next-generation firewalls - Use intrusion detection and prevention systems (IDS/IPS) - Employ network segmentation to isolate cardholder data environment - Implement web application firewalls (WAF) for e-commerce platform b) Data Protection: - Encrypt cardholder data at rest and in transit using strong cryptography (AES-256) - Implement data loss prevention (DLP) solutions - Use tokenization for payment card data where possible c) Access Control: - Implement multi-factor authentication (MFA) for all users with access to sensitive data - Use role-based access control (RBAC) - Implement the principle of least privilege - Regular access reviews and prompt termination of unnecessary access d) Vulnerability Management: - Conduct regular vulnerability scans (at least quarterly) - Perform annual penetration testing - Implement a patch management program e) Logging and Monitoring: - Centralized log management system - Security information and event management (SIEM) solution - 24/7 security operations center (SOC) f) Incident Response: - Develop and maintain an incident response plan - Conduct regular incident response drills g) Vendor Management: - Implement a vendor risk assessment program - Ensure third-party providers comply with PCI DSS and SOX requirements h) Physical Security: - Secure data centers and point-of-sale systems - Implement video surveillance and access control systems 3.

Implementation Costs (Estimated): - Network Security: \$500,000 - \$1,000,000 - Data Protection: \$300,000 - \$600,000 - Access Control: \$200,000 - \$400,000 - Vulnerability Management: \$150,000 - \$300,000 - Logging and Monitoring: \$400,000 - \$800,000 - Incident Response: \$100,000 - \$200,000 - Vendor Management: \$50,000 - \$100,000 - Physical Security: \$200,000 - \$400,000 - Staff Training and Awareness: \$100,000 - \$200,000 Total Estimated Cost: \$2,000,000 - \$4,000,000 (annual) 4.

Governance Model: a) Board of Directors: - Overall responsibility for risk management and compliance - Regular briefings on security posture and compliance status b) Executive Leadership: - CEO: Ultimate accountability for security and compliance - CFO: Oversight of financial controls and SOX compliance - CIO/CISO: Day-to-day management of security program c) Compliance Committee: - Cross-functional team including IT, Legal, Finance, and Operations - Regular meetings to review compliance status and address issues d) Internal Audit: - Independent assessment of security controls and compliance - Regular reporting to the Board's Audit Committee e) Security Operations: - Day-to-day management of security tools and processes - Incident response and threat hunting f) Training and Awareness: - Regular security awareness training for all employees - Specialized training for IT and security staff g) Policy Management: - Regular review and updates of security policies and procedures - Ensure alignment with PCI DSS and SOX requirements h) Third-Party Risk Management: - Dedicated team for assessing and monitoring vendor risks - Regular audits of critical third-party providers This security architecture provides a comprehensive approach to meeting PCI DSS and SOX compliance requirements while protecting AnyCompany's assets and reputation.

The implementation costs are estimates and may vary based on the organization's size, complexity, and existing infrastructure. Regular reviews and updates to this architecture will be necessary to address evolving threats and regulatory requirements.

5. Innovation & GenAI Opportunities

Based on the information provided, while the company profile shows many unknowns, the additional business context reveals that we are dealing with AnyCompany, a large retail and e-commerce organization. Here are some potential GenAI opportunities for AnyCompany: 1. Personalized Shopping Experiences: - Use GenAI to analyze customer behavior, purchase history, and preferences to provide highly personalized product recommendations, both online and in-store.

- Implement AI-driven chatbots for customer service that can understand and respond to complex queries, improving customer satisfaction and reducing support costs. 2. Dynamic Pricing and Inventory Management: - Utilize GenAI to predict demand fluctuations, especially during peak seasons like Black Friday, to optimize pricing strategies and inventory levels across 1,800+ locations. - Implement AI-driven forecasting to improve supply chain efficiency and reduce stockouts or overstocking. 3.

Enhanced Customer Engagement: - Develop AI-powered virtual shopping assistants that can guide customers through their shopping journey, answering questions about products and helping with size selections. - Create personalized marketing campaigns using GenAI to craft tailored messages and offers for different customer segments. 4.

Visual Search and Product Discovery: - Implement GenAI-powered visual search capabilities, allowing customers to find products by uploading images or using augmented reality features in the mobile app. 5. Fraud Detection and Security: - Use GenAI to enhance fraud detection systems, analyzing patterns in customer behavior to identify potential security threats in real-time. 6.

Content Generation: - Leverage GenAI to automatically generate product descriptions, blog posts, and social media content, ensuring consistency across all channels and reducing manual effort. 7. Voice Commerce: - Develop AI-powered voice shopping capabilities for smart home devices, allowing customers to make purchases or check inventory using voice commands. 8.

Predictive Maintenance: - Implement GenAI systems to predict maintenance needs for in-store equipment and IT infrastructure, reducing downtime and improving operational efficiency. 9. Employee Training and Support: - Create AI-driven training programs that adapt to individual employee learning speeds and styles, improving workforce productivity and knowledge retention. 10.

Customer Sentiment Analysis: - Use GenAI to analyze customer reviews, social media mentions, and support interactions to gain real-time insights into customer sentiment and identify areas for improvement. These GenAI opportunities can help AnyCompany address its key challenges, such as managing seasonal traffic spikes, improving customer experience, and enhancing operational efficiency.

Implementing these solutions could contribute significantly to AnyCompany's digital transformation goals and help maintain its competitive edge in the retail and e-commerce sector.

6. Implementation Roadmap

Based on the provided information, I'll design a comprehensive AWS Landing Zone for AnyCompany, a large retail organization with 1,800+ locations nationwide. This design will support their digital transformation initiative and address their specific needs for scalability, security, and compliance. AWS Landing Zone Design for AnyCompany: 1.

Account Structure: a) Management Account - Purpose: Centralized billing and AWS Organizations management - Key services: AWS Organizations, AWS SSO, AWS Config b) Security Account - Purpose: Centralized security controls and monitoring - Key services: AWS Security Hub, GuardDuty, CloudTrail, Config c) Shared Services Account - Purpose: Common resources and services used across the organization - Key services: Transit Gateway, Route 53, Certificate Manager d) Log Archive Account - Purpose: Centralized log storage and analysis - Key services: CloudWatch Logs, S3, Athena e) Production Account - Purpose: Production workloads for e-commerce platform - Key services: EC2, ECS, RDS, ElastiCache, S3 f) Development/Test Account - Purpose: Development and testing environments - Key services: EC2, ECS, RDS, ElastiCache, S3 g) Data Analytics Account - Purpose: Big data processing and analytics - Key services: EMR, Redshift, Athena, QuickSight h) Disaster Recovery Account - Purpose: Backup and disaster recovery resources - Key services: S3, EBS snapshots, RDS snapshots, DynamoDB backups 2.

Network Architecture: a) VPC Design: - Implement a multi-VPC architecture with separate VPCs for production, development, and shared services - Use VPC peering or Transit Gateway for inter-VPC communication b) Subnets: - Public subnets for internet-facing resources - Private subnets for application and database tiers - Implement network ACLs and security groups for granular access control c) Connectivity: - Use AWS Direct Connect for high-bandwidth, low-latency connectivity to on-premises data centers - Implement Site-to-Site VPN as a backup and for smaller locations - Use AWS Global Accelerator for improved performance across regions d) Content Delivery: - Implement Amazon CloudFront for global content delivery and DDoS protection 3.

Security Controls: a) Identity and Access Management: - Use AWS SSO for centralized user management - Implement fine-grained IAM policies following the principle of least privilege - Enable MFA for all IAM users b) Data Protection: - Encrypt data at rest using AWS KMS - Use AWS Certificate Manager for managing SSL/TLS certificates - Implement AWS Macie for sensitive data discovery and protection c) Network Security: - Use AWS Network Firewall for centralized network protection - Implement AWS WAF for web application protection - Enable VPC Flow Logs for network traffic analysis d) Compliance: - Use AWS Config for continuous compliance monitoring - Implement AWS Security Hub for security posture management -

Use AWS Audit Manager for streamlined compliance audits 4.

Governance Framework: a) Policy Management: - Use Service Control Policies (SCPs) to enforce organization-wide policies - Implement AWS Config Rules for continuous compliance checks b) Cost Management: - Use AWS Budgets for cost allocation and tracking - Implement tagging strategies for cost attribution c) Resource Management: - Use AWS Resource Access Manager for sharing resources across accounts - Implement AWS Service Catalog for standardized resource provisioning 5.

Operational Tools: a) Monitoring and Logging: - Centralize logs in the Log Archive account using CloudWatch Logs - Use Amazon EventBridge for event-driven architecture b) Automation: - Implement AWS Systems Manager for patch management and automation - Use AWS Lambda for serverless automation tasks c) Deployment: - Use AWS CodePipeline and CodeDeploy for CI/CD - Implement Infrastructure as Code using AWS CloudFormation or Terraform 6.

Scalability and Performance: a) Auto Scaling: - Implement EC2 Auto Scaling groups for the e-commerce platform - Use Application Auto Scaling for services like RDS and DynamoDB b) Load Balancing: - Use Elastic Load Balancing (ALB/NLB) for distributing traffic c) Caching: - Implement Amazon ElastiCache for application caching - Use DAX for DynamoDB caching 7.

Disaster Recovery: a) Backup: - Use AWS Backup for centralized backup management - Implement cross-region replication for critical S3 buckets b) Recovery: - Design multi-region architectures for critical applications - Use Route 53 for DNS failover This AWS Landing Zone design provides AnyCompany with a secure, scalable, and compliant foundation for their cloud journey.

It addresses their need for handling seasonal traffic spikes, improving customer experience, and modernizing their infrastructure. The design can be further refined based on specific application requirements and future business needs.